

TEZCAT ARCHITECTURE

deterministic laboratory | simulation ≠ forecasting

RESEARCH INTERFACES via CONTROL PLANE

**Researcher
CLI / dashboard**

React/Vite dashboard
Build · Research · Lab · Markets · Ops

**FastAPI control plane
/api/***

Registry · run manager · research API
TradeOps queue

TEZCAT ECOLOGY ENGINE | deterministic step loop

ENVIRONMENT
SHOCKS: whale · MM pull
REGIMES: stable · crisis · recovery

AGENTS + MEMORY
5 heterogeneous types
Noise · retail · momentum
Memory: fear · confidence

LIMIT ORDER BOOK
Price-time priority
Tick snapping · partial fills
Matching engine

PORTFOLIO + RISK
Reservations · margin
Liquidation as order flow
No negative balances

Event log | hash chain
Replay · checkpoints · forks

Metrics | returns · spread · depth
imbalance · drawdown

**Endogenous price
no external series in normal runs**

RESEARCH + PROVENANCE LAYER

ExperimentVersion
canonical JSON · seed · SHA-256

BatchRunner / TradeOps
replicate · resume · serial ≡ parallel

Analysis / calibration
bootstrap · permutation · OOS

Reports / replay
artifacts · manifests · reproduce

EXTERNAL DATA | read-only

Kalshi
datasets

Polymarket
signatures

SYNTHETIC WORLDS

world hash · ecology fingerprint
quote / trade stream

NAUTILUS | BACKTEST ONLY

QuoteTick /
TradeTick

Strategies → metrics
observer only

RESEARCH CONTROL PLANE (S5)

dataset → forecast → portfolio → world → backtest → risk → report
hash-linked reproduction

DEVOPS STATUS

IN REPO: Docker · Compose · Actions · Pages
Render · **AWS SAM**
FUTURE: Daytona · Kubernetes · Kind · Jenkins

— solid — implemented path

- - - dashed purple - - - external observation